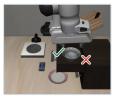
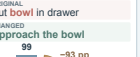
a Motivation

Semantic following



ORIGINAL
Put **bowl** in drawer
CHANGED
Approach the bowl



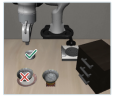
Multimodal instructions



ORIGINAL
Put bowl **on plate**
CHANGED
Put bowl in **red circle**



Mid-execution correction



ORIGINAL
Put bowl **on plate**
UPDATED
No, don't put the bowl on the plate. Put it in the ramekin instead.

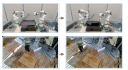


$\pi_{0.5}$: diagnostic instruction completion (%)

High task success does not necessarily mean a *'smart'* policy.

b Data construction

Demonstrations



Slice the lotus root with a knife.

145,179 retained trajectories

858,915 action segments





SmartVLA

c Policy & training

SmartVLA-OFT

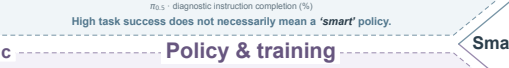
1 pass



SmartVLA-Pi

flow × N





d Evaluation

LIBERO-SMART

120 tasks
6,000 instances

Strict success (%) $\pi_{0.5}$ 7.18
SmartVLA-Pi 14.67
SmartVLA-OFT 14.03
0 10 20

SEMANTIC
Paraphrases
Subtasks
Atomic actions
Logical constraints

VISUAL PROMPT
Visual target selection
Target localization

CONTROL
Goal correction
Process replanning

Benchmark results

